



Ezeddin Al Hakim

Principal AI Engineer

Languages English | Swedish | Arabic

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SUMMARY

I'm a builder with deep, wide-spectrum experience across AI—from research and invention to engineering and large-scale infrastructure. I've worked on LLMs, edge AI, robotics, telecom, and autonomous systems at Samsung (Chief AI Engineer), Ericsson, Scania, and Vodafone. Currently Head of AI Ops & Infrastructure at Humanoid, I support 80+ engineers and lead scalable AI systems. I'm also the founder of Swibit, delivering private AI solutions across EMEA. My work includes 26 patents, 10+ papers, and an M.Sc. in AI from ETH Zurich and KTH Stockholm.

WORK EXPERIENCE

- Present**  **Humanoid | Head of AI Ops & Infrastructure | United Kingdom**
05/2024
Building advanced humanoid robots designed for seamless integration into daily life. Oversee DevOps, MLOps, and DataOps operations, supporting over 80 engineers daily across AI, software, and robotics teams.
- Key Technologies Used**
MLOps | DevOps | DataOps | LLM | GenAI | Leadership | Infrastructure | Kubernetes | MLOps | CI/CD
- Present**  **Founder at Swibit | Co-Founder at AgileLoop | United Kingdom**
09/2023
Founded Swibit and co-founded AgileLoop, AI consulting and solutions companies serving the EMEA region. Led multiple large-scale contracts with large organisation and enterprise clients, delivering secure on-perm LLMs, vision AI, and AI-powered ERP systems.
- Personal Contract via Swibit**
- **Head of AI, UK:** Led the design and deployment of an SDK enabling 100% on-device LLMs for iOS and Android apps—supporting offline use, data privacy, and cost-effective inference. Drove cross-industry adoption and unlocked new B2B use cases.
 - **Principal AI Engineer, UK:** Developed interactive digital human simulations using video, voice, and text based on user data. Built a scalable micro-service architecture and deployed self-hosted LLM models on AWS using LLMops and CI/CD best practices.
- 09/2023**  **Samsung | Staff / Chief AI Engineer | United Kingdom**
03/2022
Samsung AI Research, London. Worked in cutting edge research and proof-of-concept development within AI technologies to automate 5G and 6G Mobile Networks.
- Achievements/Tasks**
- Development of MLOps for 6G networks, including data management, model development and configuration, and model deployment.
 - Development of AI solutions for 6G networks, including LLM for mobile network, AI Conflict Management and Trustworthy AI
 - Awarded the Samsung Inspire Award 2023 for taking action and finding solutions.
 - More than 12 patent applications and 2 academic publications have been filed.
- Key Technologies Used**
MLOps | LLM | Reinforcement Learning | Transfer Learning | Python | Tensorflow | PyTorch | Ray | RLlib | Kubernetes | Docker
- 03/2022**  **Vodafone Group | Data Scientist | United Kingdom**
10/2021
Vodafone Group HQ, London. Develop RL recommender systems for use across multiple countries, which involved large and varied datasets.
- Key Technologies Used**
Python | Reinforcement Learning | Tensorflow | PyTorch | Google Cloud | Kubernetes | Docker | AirFlow | MLOps | CI/CD
- 10/2021**  **Ericsson | AI Experienced Researcher | Sweden**
11/2019
Ericsson AI Research. Worked in cutting edge research and proof-of-concept development within AI technologies to automate 5G and 6G Mobile Networks. Worked in close cooperation with global customers, Ericsson business units and academic partners.
- Achievements/Tasks**
- Awarded the Ericsson CEO's Key Contributor Award for 2021.
 - Research achievements include: 13 patent applications, academic publications and 10 master's thesis supervisions.
 - Lead AI projects (4-10 people) focused on reinforcement learning, transfer learning, and distributed learning in network optimization.
 - Developed AI pipelines for live mobile networks, including data management, distributed training, and cloud deployment.
 - Developed a framework, tools, and library to be used across Ericsson AI Research's projects.
- Key Technologies Used**
Reinforcement Learning | Federated Learning | Sim2Real | Generative AI | Python | Tensorflow | PyTorch | Ray | RLlib | Kubernetes | Docker
- 11/2019**  **Scania | AI Development Engineer | Sweden**
01/2018
Scania R&D, Autonomous Systems. Developed AI perception solutions for autonomous vehicle perception in collaboration with the Volkswagen Group (VW, Audi, and Porsche). ([demo video](#)).
- Achievements/Tasks**
- A novel neural network, 3D YOLO, was developed to detect and classify 3D objects, such as vehicles and pedestrians, in real-time from LIDAR data. As a result of its real-time efficiency, it has proven appealing in the field of autonomous vehicles around the world.
 - Computer vision models have been developed for automated driving with real-time performance in complex environments.
- Key Technologies Used**
Deep Learning | Computer Vision | Python | C++ | Tensorflow | AWS | Docker | Object detection | Semantic Segmentation | 3D Point Cloud

EDUCATION

- 02/2018
01/2017  **Swiss Federal Institute of Technology (ETH Zürich) | Zürich, Switzerland**
M.S.c in Machine Learning & Software Engineering
An exchange year programme from Royal Institute of Technology (KTH) in Stockholm. Accepted with a Swedish GPA of 4.5 / 5.
Relevant Courses
Big Data | Natural Language Processing (NLP, LSTM, RNN) | Theory of Robotics | Fundamentals of Mathematical Statistics | Machine Learning and Matrix Factorisation | Smart Wearable Systems | Autonomous Mobile Robots | Statistical Learning Theory
- 06/2018
08/2013  **Royal Institute of Technology (KTH) | Stockholm, Sweden**
M.S.c in Machine Learning & Software Engineering
A two-year programme of 120 ECTS providing the mathematical and statistical foundations and methods for Machine Learning with the goal of modelling and discovering patterns from observations.
Relevant Courses
Machine Learning (SVM, DNN, K-Means, Random Forest, Regression, PCA, etc) | Artificial Intelligence | An advanced course in Machine Learning | Image Based Recognition and Classification (Deep Learning) | Search Engines and Information Retrieval Systems
- B.S.c in Computer Science & Software Engineering**
A three-year programme of 180 ECTS offers courses on programming, algorithms, data structures, and analysis for reliable software systems.
Relevant Courses
Operating Systems (C Lang) | Algorithms, Data Structures and Complexity (Java) | Computer Organization and Components (C lang, Assembly) | Programming Paradigms (Haskell, Erlang, Prolog, COBOL, etc.) | Database Technolog | Numerical Mathematics (MATLAB)
- 06/2012
08/2009  **High School - IT-Gymnasiet | Stockholm, Sweden**
Computer Science
Relevant Courses
Programming A (Python) | Programming B (Ruby) | Programming C (VS Basic, .Net C#) | Database Management (SQL) | Network Security

SELECTED PATENTS

- 02/2022  **WO2022028926A1: Offline Simulation-to-Reality Transfer for Reinforcement Learning (RL Sim2Real)**
- 02/2022  **WO2022023218A1: Methods and Apparatus for Managing a System that Controls an Environment (Safe AI/RL)**
- 03/2022  **WO2022060264A1: Methods and Systems for Updating Machine Learning Models (Federated Learning)**
- 09/2022  **WO2022188957A1: Methods and apparatus for managing an environment within a domain (Safe AI/RL)**
- 12/2022  **WO2022258846A1: Systems and methods for machine-learned optimization of wireless networks (Graph Attention Networks)**

SELECTED PUBLICATIONS

- 12/2021  **A Graph Attention Learning Approach to Antenna Tilt Optimization**
IEEE (Preprint arXiv): <https://arxiv.org/abs/2112.14843>
- 12/2020  **Dynamic Clustering in Federated Learning**
IEEE (Preprint arXiv): <https://arxiv.org/abs/2012.03788>
- 12/2020  **Safe Reinforcement Learning for Antenna Tilt Optimisation using Shielding and Baselines**
IEEE (Preprint arXiv): <https://arxiv.org/abs/2012.01296>
- 10/2020  **Remote Electrical Tilt Optimization via Safe Reinforcement Learning**
IEEE (Preprint arXiv): <https://arxiv.org/abs/2010.05842>
- 08/2018  **3D YOLO: End-to-End 3D Object Detection Using Point Clouds**
DiVA (Thesis Report): <http://www.diva-portal.org/smash/record.jsf?pid=diva2:1245296>

SKILLS

Software (advanced)	Python: [Tensorflow Keras PyTorch NumPy Scikit OpenCV Pandas RLLib] Java
Software (basic)	C/C++ C# MATLAB JavaScript Node.js SQL NoSQL Ray Hadoop Git Unix
Domains	Artificial Intelligence Deep Learning Computer Vision Reinforcement Learning Big Data
Other	Amazon Web Services (AWS) Docker Kubernetes Azure Jira Agile Web design

HOBBIES & INTERESTS

Photography | Traveling | History | Design | Solving Programming Exercises